# Jiayi Li

**Unreal Engine C++ Developer** | **Gameplay & Combat Systems**

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## Unreal Engine Projects

**Multiplayer FPS** In Development

- Competitive multiplayer FPS developed in C++ with Unreal Engine 5.8, supporting first- and third-person character presentation on dedicated, listen, and standalone servers.
- Built a data-driven weapon pipeline by storing weapon behavior and presentation in Data Assets referenced through Gameplay Tags, enabling designers to add or tune weapons without modifying core C++ code.
- Designed server-authoritative combat by predicting fire locally and reconciling ammunition, damage, impacts, and effects from validated server state, delivering responsive and consistent multiplayer feedback.

**Shooter Sam**

- Single-player third-person shooter developed with C++ and Blueprints in Unreal Engine 5.6, centered on tactical AI and objective-driven levels.
- Designed patrol, investigation, pursuit, and attack behavior with Behavior Tree services, decorators, and visibility traces, enabling clean state transitions while preventing enemies from tracking the player through walls.
- Implemented squad awareness by converting gunfire into shared last-known position and direction data, allowing nearby AI to coordinate an investigation before resuming individual decisions.
- Built an event-driven objective framework in which collection, elimination, and key events update GameMode state, unlocking the exit only after all requirements are satisfied.
- Developed an animation state machine from player input, velocity, movement direction, and actor rotation, producing responsive transitions between locomotion, aiming, and attack states.
- Created a reusable C++ health component that handles damage and broadcasts normalized health through a multicast delegate, allowing player and AI UMG health bars to update without Tick polling.
- Optimized AI updates by profiling workload and using octree queries, visibility traces, and NavMesh connectivity checks to suspend Tick for unreachable enemies, reducing unnecessary processing.

**Battle Blaster**

- Single-player tank shooter developed with C++ and Blueprints in Unreal Engine 5.6, featuring vehicle movement, turret aiming, projectile combat, and enemy towers.
- Designed a shared pawn combat architecture by extracting turret rotation, projectile spawning, and damage handling into reusable logic, enabling tank and tower variants without duplicating core systems.
- Implemented tank movement by converting throttle and steering input into swept translation and rotation against a box-based collision root, producing responsive controls while preventing tunneling and invalid overlaps.
- Built the tower aiming pipeline by calculating target direction, interpolating turret rotation at a constant rate, and gating projectile spawning with timers, producing predictable tracking and firing cadence.

## Graphics & Rendering Projects

- **Custom Rendering Engine:** Built a C++/OpenGL engine by combining VAO/VBO rasterization with ray and path tracing, supporting interactive scene previews and higher-quality offline output.
- **Lighting & Materials:** Implemented direct and global illumination through attenuated lights, shadows, recursive reflection and refraction, UV textures, glossy materials, and supersampling, improving light and material response.
- **Adaptive Geometry:** Adjusted tessellation from camera distance and implemented subdivision, simplification, and remeshing tools, increasing nearby detail while controlling geometric complexity.
- **Fishing Game:** Combined procedural terrain, dynamic weather, physically based sky rendering, and animated water in the custom engine, creating a fluid and natural outdoor fishing environment.

## Education

**Brown University** – Providence, RI May 2026

*Master of Science in Computer Science*

**Boston University** – Boston, MA Jan 2024

*Bachelor of Arts in Computer Science and Statistics*